

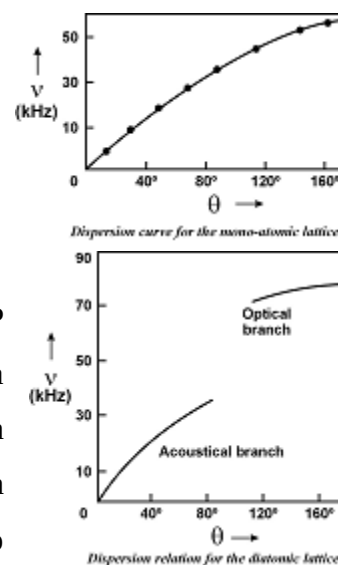
LATTICE DYNAMICS KIT[®]

OBJECTIVE

1. *STUDY of the Dispersion relation for “Mono-atomic Lattice”. Determination of the Cut-off frequency and Comparison with theory.*
2. *Study of the Dispersion relation for the Di-atomic Lattice, Acoustical mode and Energy Gap. Comparison with theory.*



Lattice dynamics is an essential component of any postgraduate course in Physics, Engineering Physics, Electronic Engineering and Material Science. In particular it is essential to understanding the interaction of electromagnetic waves and crystalline solids. In general, students find it difficult to understand involved concepts like ACOUSTICAL MODE, OPTICAL MODE, and ENERGY GAP etc, which they cannot see for themselves in the laboratory. Such a difficulty can be overcome by introducing a laboratory exercise in which the student follows a carefully prescribed procedure, which presents him with a simplified model of the system and allows him to verify well-established theories. In the process he gains an insight into the concepts. The lattice dynamic Kit provides such an experience in the study of mono and di-atomic lattices.



INSTRUMENT It consists of an Audio oscillator with amplitude control and facility to vary the frequency from 0.9 KHz to 90 KHz. It has built in powersupply and output stage to match the impedance of simulated lattice. Another part of Lattice Dynamic Kit consists of transmission line, which simulates one-dimensional mono-atomic and di-atomic lattices. The only additional equipment needed is a General purpose C.R.O.

Manufacturers:



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